Tugas Personal 1

**Perancangan & Implementasi Database**

 Logo MAPAN

# Sistem Informasi MAPAN

Manajemen & Analisis Prediksi Agri Nusantara

Analisis sistem · ERD · CDM/PDM · SQL DDL & DML · pengujian query

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# 1. Analisis Sistem Informasi

## 1.1 Latar belakang

Kelompok tani/gapoktan umumnya mencatat kegiatan pertanian secara manual (buku tulis, WhatsApp, dan ingatan pengurus). Data lahan, komoditas, anggota, dan hasil panen terserak dan cepat hilang, sehingga laporan ke penyuluh/dinas selalu memakan waktu dan keputusan musim berikutnya dilakukan tanpa dasar data. **MAPAN** dibangun sebagai sistem informasi web untuk menyatukan data tersebut dalam satu ruang kerja organisasi.

## 1.2 Tujuan sistem

1. Menjadikan satu sumber data tunggal untuk lahan, anggota, dan kegiatan.
2. Menghitung ringkasan (KPI): jumlah lahan, luas total, jumlah dan progres kegiatan.
3. Memisahkan hak akses: administrator, manajer, petani, penyuluh (baca saja).
4. Menghasilkan laporan siap unduh (CSV) untuk pelaporan organisasi/dinas.

## 1.3 Kebutuhan data

| Kebutuhan | Data yang dibutuhkan |
|---|---|
| Identitas organisasi & anggota | Organisasi, pengguna, keanggotaan dengan peran |
| Inventaris lahan | Nama lahan, luas, status kesehatan, komoditas, petani pengelola |
| Kegiatan lapangan | Catatan aktivitas/tugas/panen/scan, tanggal, progres |
| Master produk | Komoditas, kategori, benchmark produktivitas (ton/ha) |

## 2. Perancangan ERD

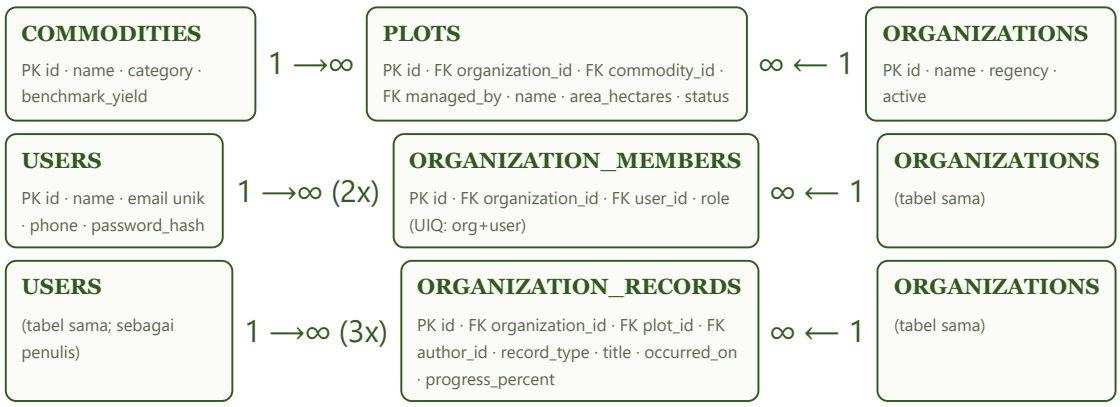
Tahap ini mengidentifikasi entitas (objek data utama), atributnya, kunci utama (*primary key*), serta relasi antar entitas. Enam entitas berikut dipilih sebagai **subset nyata dari database MAPAN**: hanya bagian yang paling mewakili fungsi utama sistem yang diambil, agar memenuhi ketentuan minimal lima tabel berelasi tanpa melepas konteks proyek.

### 2.1 Entitas, atribut, dan primary key

| Entitas | Primary key | Atribut penting |
|---|---|---|
| COMMODITIES | `id` | name, category, benchmark_yield |
| USERS | `id` | name, email (unik), phone, password_hash |
| ORGANIZATIONS | `id` | name, regency, active |
| ORGANIZATION_MEMBERS | `id` | organization_id (FK), user_id (FK), role, joined_at — (organization_id, user_id) unik |
| PLOTS | `id` | organization_id (FK), commodity_id (FK), managed_by (FK), name, area_hectares, status |
| ORGANIZATION_RECORDS | `id` | organization_id (FK), plot_id (FK), author_id (FK), record_type, title, occurred_on, progress_percent |

### 2.2 Diagram ERD (notasi “kaki gagak” / crow's foot)

Cara membaca: kotak = entitas/tabel; tulisan di dalamnya = atribut (PK = kunci utama, FK = kunci asing/tamu); simbol panah menunjukkan arah relasi dan kardinalitasnya (satu — banyak). Tanda (2x)/(3x) berarti tabel USERS ikut terlibat dalam dua dan tiga relasi berbeda.




| Relasi | Jenis | Keterangan bisnis |
|---|---|---|
| ORGANIZATIONS 1—∞ ORGANIZATION_MEMBERS ∞—1 USERS | N : N | Satu anggota bisa ikut beberapa organisasi dengan peran berbeda |
| ORGANIZATIONS 1—∞ PLOTS | 1 : N | Satu organisasi punya banyak lahan |
| COMMODITIES 1—∞ PLOTS | 1 : N | Satu komoditas ditanam di banyak lahan |
| USERS 1—∞ PLOTS (managed_by) | 1 : N | Satu petani mengelola beberapa lahan |
| ORGANIZATIONS 1—∞ ORGANIZATION_RECORDS | 1 : N | Catatan selalu milik satu organisasi (ciri aplikasi multi-organisasi) |

| | | |
|---|---|---|
| PLOTS 0..1—∞ ORGANIZATION_RECORDS | opsional 1 : N | Catatan boleh tidak terkait lahan spesifik |
| USERS 1—∞ ORGANIZATION_RECORDS (author) | 1 : N | Setiap catatan memiliki penulis |

## 3. Perancangan Database — CDM dan PDM

Bagian ini menerjemahkan hasil rancangan tingkat konsep (CDM — *Conceptual Data Model* / Model Data Konseptual: gambaran entitas dan relasi tanpa detail teknis) menjadi rancangan tingkat implementasi (PDM — *Physical Data Model* / Model Data Fisik: tipe data, kunci, batasan, dan mesin basis datanya). Tabel berikut memetakan kedua tingkat tersebut dari ERD pada Bab 2:

| Aspek | CDM (konseptual) | PDM (fisik — MySQL 8) |
|---|---|---|
| Entitas → objek | Entitas COMMODITIES, USERS, ORGANIZATIONS, ORGANIZATION_MEMBERS, PLOTS, ORGANIZATION_RECORDS | Tabel InnoDB `utf8mb4_unicode_ci` |
| Kunci | Identifier (pembeda): id pada tiap entitas | `INT AUTO_INCREMENT PRIMARY KEY` |
| Relasi | Keterkaitan antar entitas (1—N, N—N) | `FOREIGN KEY (...) REFERENCES ... (id)` + `UNIQUE KEY` untuk N:N (members) |
| Atribut kuantitas | luas lahan, benchmark | `DECIMAL(8,3)`, `DECIMAL(6,2)` |
| Atribut status/tipe | status lahan; peran; jenis catatan | `ENUM('sehat','waspada','kritis')`, `ENUM('pangan','hortikultura','perkebunan')`, `ENUM('administrator','manager','farmer','extension_officer')`, `ENUM('activity','task','harvest','scan')` |
| Data lain | tanggal kejadian, progres | `DATE`, `TINYINT 0–100` |
| Keunikan | email pengguna unik; satu orang satu peran per organisasi | `UNIQUE` pada `email`; `UNIQUE KEY uq_member(organization_id, user_id)` |
| Kolom audit | waktu dibuat | `TIMESTAMP DEFAULT CURRENT_TIMESTAMP` |

## 4. Implementasi Database — SQL DDL

Skrip lengkap: `mapan-tugas1.sql` (file terlampir; di bawah cuplikannya). Database dibuat terpisah agar aman dan bisa diulang:

```
CREATE DATABASE mapan_tugas1 CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE mapan_tugas1;

CREATE TABLE commodities (
  id              INT AUTO_INCREMENT PRIMARY KEY,
  name            VARCHAR(80)  NOT NULL,
  category        ENUM('pangan','hortikultura','perkebunan') NOT NULL,
  benchmark_yield DECIMAL(6,2) NOT NULL,
  created_at      TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE users (
  id            INT AUTO_INCREMENT PRIMARY KEY,
  name          VARCHAR(100) NOT NULL,
  email         VARCHAR(150) NOT NULL UNIQUE,
  phone         VARCHAR(20)  NULL,
  password_hash CHAR(60)     NOT NULL,
  created_at    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE organizations (
  id         INT AUTO_INCREMENT PRIMARY KEY,
  name       VARCHAR(150) NOT NULL,
  regency    VARCHAR(100) NOT NULL DEFAULT 'Kab. Semarang',
  active     TINYINT(1)   NOT NULL DEFAULT 1,
  created_at TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE organization_members (
  id              INT AUTO_INCREMENT PRIMARY KEY,
  organization_id INT NOT NULL,
  user_id         INT NOT NULL,
  role            ENUM('administrator','manager','farmer','extension_officer') NOT NULL,
  joined_at       TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  UNIQUE KEY uq_member (organization_id, user_id),
  FOREIGN KEY (organization_id) REFERENCES organizations (id),
  FOREIGN KEY (user_id)         REFERENCES users (id)
);

CREATE TABLE plots (
  id              INT AUTO_INCREMENT PRIMARY KEY,
  organization_id INT NOT NULL,
  commodity_id    INT NOT NULL,
  managed_by      INT NULL,
  name            VARCHAR(120) NOT NULL,
  area_hectares   DECIMAL(8,3) NOT NULL,
  status          ENUM('sehat','waspada','kritis') NOT NULL DEFAULT 'sehat',
  FOREIGN KEY (organization_id) REFERENCES organizations (id),
  FOREIGN KEY (commodity_id)    REFERENCES commodities (id),
  FOREIGN KEY (managed_by)      REFERENCES users (id)
);

CREATE TABLE organization_records (
  id               INT AUTO_INCREMENT PRIMARY KEY,
  organization_id  INT NOT NULL,
  plot_id          INT NULL,
```

```
  author_id        INT NOT NULL,
  record_type      ENUM('activity','task','harvest','scan') NOT NULL,
  title            VARCHAR(160) NOT NULL,
  occurred_on      DATE         NOT NULL,
  progress_percent TINYINT      NOT NULL DEFAULT 0,
  FOREIGN KEY (organization_id) REFERENCES organizations (id),
  FOREIGN KEY (plot_id)         REFERENCES plots (id),
  FOREIGN KEY (author_id)       REFERENCES users (id)
);
```

# 5. Pengisian Data — SQL DML

## 5.1 INSERT (pengisian simulasi)

```
INSERT INTO commodities (name, category, benchmark_yield) VALUES
  ('Padi','pangan',5.20), ('Jagung','pangan',5.80), ('Cabai','hortikultura',8.00),
  ('Tomat','hortikultura',25.00), ('Kopi','perkebunan',1.10), ('Ubi','pangan',9.00);

INSERT INTO users (name, email, phone, password_hash) VALUES
  ('Siti Rahayu','siti@tani.id','081200000001','$2y$10$hashsimulasitugas1...0001'),
  ('Budi Santoso','budi@tani.id','081200000002','$2y$10$hashsimulasitugas1...0002'),
  ('Agus Priono','agus@tani.id','081200000003','$2y$10$hashsimulasitugas1...0003'),
  ('Rina Wulandari','rina@tani.id','081200000004','$2y$10$hashsimulasitugas1...0004'),
  ('Joko Susilo','joko@tani.id','081200000005','$2y$10$hashsimulasitugas1...0005'),
  ('Dewi Astuti','dewi@tani.id','081200000006','$2y$10$hashsimulasitugas1...0006');

INSERT INTO organizations (name, regency) VALUES
  ('Gapoktan Tani Makmur','Kab. Semarang'), ('Koperasi Tani Harapan','Kab. Semarang'),
  ('Gapoktan Mekar Jaya','Kab. Temanggung'), ('Kelompok Hortikultura Sari','Kab. Boyolal
  ('Koperasi Kopi Buana','Kab. Temanggung');

INSERT INTO organization_members (organization_id, user_id, role) VALUES
  (1,1,'administrator'), (1,2,'manager'), (1,3,'farmer'), (1,4,'extension_officer'),
  (2,5,'administrator'), (2,3,'farmer'), (3,6,'administrator'), (4,1,'manager');

INSERT INTO plots (organization_id, commodity_id, managed_by, name, area_hectares, statu
  (1,1,3,'Sawah Blok Utara',1.250,'sehat'), (1,2,3,'Jagung Blok Selatan',0.800,'waspada'
  (1,3,2,'Kebun Cabai Belakang',0.400,'kritis'), (2,4,5,'Kebun Tomat Rini',0.500,'sehat'
  (2,1,5,'Sawah Debogor',2.100,'sehat'), (3,5,6,'Lahan Kopi Lereng',3.500,'waspada');

INSERT INTO organization_records (organization_id, plot_id, author_id, record_type, titl
  (1,1,3,'activity','Pemupukan tahap 2 Sawah Blok Utara','2026-10-05',100),
  (1,2,3,'task','Penyemprotan hama Jagung Selatan','2026-10-12',70),
  (1,3,2,'scan','Pemeriksaan gejala daun keriting cabai','2026-10-15',60),
  (1,1,1,'harvest','Panen perdana Sawah Blok Utara','2026-10-30',100),
  (2,5,5,'activity','Penyiaran benih Sawah Debogor','2026-09-28',100),
  (2,4,5,'harvest','Panen tomat gelombang 1','2026-10-20',100),
  (3,6,6,'activity','Pemangkasan kopi Lereng','2026-10-08',45);
```

Ringkasan jumlah baris per tabel (dari hasil eksekusi): **commodities 6 · users 6 · organizations 5 · organization_members 8 · plots 6 · organization_records 7** — memenuhi syarat min. 5 baris per tabel.

## 5.2 UPDATE (manipulasi pembaruan)

```
UPDATE plots SET status='waspada' WHERE id=3;                  -- lahan kritis sudah dit
UPDATE organization_records SET progress_percent=100
  WHERE title='Penyemprotan hama Jagung Selatan';              -- tugas selesai
```

## 5.3 DELETE (manipulasi penghapusan)

```
INSERT INTO organization_records (organization_id, plot_id, author_id, record_type, titl
  VALUES (1,2,3,'task','Catatan sementara untuk dihapus','2026-10-18');
DELETE FROM organization_records WHERE title='Catatan sementara untuk dihapus';
-- hasil eksekusi: 1 row affected (baris simulasi dibuat lalu dihapus)
```

## 6. Pengujian Query + Hasil Eksekusi Nyata

Aktivitas eksekusi: MySQL 8.4.3 lokal, user `mapan`, database `mapan_tugas1`. Bukti eksekusi utuh: file `hasil-eksekusi.txt` (transkrip CLI, 14 result set, tanpa ERROR).

### Q1 — Menampilkan data (SELECT + 2 JOIN)

```
SELECT p.id, p.name AS lahan, c.name AS komoditas, o.name AS organisasi,
       p.area_hectares AS luas_ha, p.status
FROM plots p
JOIN commodities c   ON c.id = p.commodity_id
JOIN organizations o ON o.id = p.organization_id;
```

```
+----+----------------------+-----------+----------------------+---------+---------+
| id | lahan                | komoditas | organisasi           | luas_ha | status  |
+----+----------------------+-----------+----------------------+---------+---------+
|  1 | Sawah Blok Utara     | Padi      | Gapoktan Tani Makmur |   1.250 | sehat   |
|  2 | Jagung Blok Selatan  | Jagung    | Gapoktan Tani Makmur |   0.800 | waspada |
|  3 | Kebun Cabai Belakang | Cabai     | Gapoktan Tani Makmur |   0.400 | waspada |
|  4 | Kebun Tomat Rini     | Tomat     | Koperasi Tani Harapan |   0.500 | sehat   |
|  5 | Sawah Debogor        | Padi      | Koperasi Tani Harapan |   2.100 | sehat   |
|  6 | Lahan Kopi Lereng    | Kopi      | Gapoktan Mekar Jaya  |   3.500 | waspada |
+----+----------------------+-----------+----------------------+---------+---------+
6 rows in set (0.00 sec)
```

### Q2 — Menggabungkan 3 tabel (users · members · organizations)

```
SELECT u.name AS pengguna, u.email, m.role AS peran, o.name AS organisasi
FROM organization_members m
JOIN users u         ON u.id = m.user_id
JOIN organizations o ON o.id = m.organization_id
ORDER BY o.name, m.role;
```

```
+----------------+--------------+-------------------+----------------------------+
| pengguna       | email        | peran             | organisasi                 |
+----------------+--------------+-------------------+----------------------------+
| Dewi Astuti    | dewi@tani.id | administrator     | Gapoktan Mekar Jaya        |
| Siti Rahayu    | siti@tani.id | administrator     | Gapoktan Tani Makmur       |
| Budi Santoso   | budi@tani.id | manager           | Gapoktan Tani Makmur       |
| Agus Priono    | agus@tani.id | farmer            | Gapoktan Tani Makmur       |
| Rina Wulandari | rina@tani.id | extension_officer | Gapoktan Tani Makmur       |
| Siti Rahayu    | siti@tani.id | manager           | Kelompok Hortikultura Sari |
| Joko Susilo    | joko@tani.id | administrator     | Koperasi Tani Harapan      |
| Agus Priono    | agus@tani.id | farmer            | Koperasi Tani Harapan      |
+----------------+--------------+-------------------+----------------------------+
8 rows in set (0.00 sec)
```

### Q3a — Menyaring data (WHERE + OR)

```
SELECT p.name AS lahan, o.name AS organisasi, p.area_hectares AS luas_ha, p.status
FROM plots p
JOIN organizations o ON o.id = p.organization_id
WHERE p.status = 'kritis' OR p.area_hectares > 1.000;
```

```
+------------------+----------------------+---------+---------+
| lahan            | organisasi           | luas_ha | status  |
+------------------+----------------------+---------+---------+
| Sawah Blok Utara | Gapoktan Tani Makmur |   1.250 | sehat   |
| Sawah Debogor    | Koperasi Tani Harapan |   2.100 | sehat   |
| Lahan Kopi Lereng | Gapoktan Mekar Jaya  |   3.500 | waspada |
```

```
+------------------+----------------------+---------+---------+
3 rows in set (0.00 sec)
```

## Q3b — Menyaring (panen selesai, LEFT JOIN)

```
SELECT r.title, r.occurred_on, r.progress_percent, p.name AS lahan
FROM organization_records r
LEFT JOIN plots p ON p.id = r.plot_id
WHERE r.record_type = 'harvest' AND r.progress_percent >= 100;
```

```
+--------------------------------+-------------+------------------+------------------+
| title                          | occurred_on | progress_percent | lahan            |
+--------------------------------+-------------+------------------+------------------+
| Panen perdana Sawah Blok Utara | 2026-10-30  |              100 | Sawah Blok Utara |
| Panen tomat gelombang 1        | 2026-10-20  |              100 | Kebun Tomat Rini |
+--------------------------------+-------------+------------------+------------------+
2 rows in set (0.00 sec)
```

## Q4a — Merekapitulasi (GROUP BY + COUNT + SUM)

```
SELECT o.name AS organisasi, COUNT(*) AS jumlah_lahan,
       SUM(p.area_hectares) AS total_luas_ha
FROM plots p
JOIN organizations o ON o.id = p.organization_id
GROUP BY o.name;
```

```
+----------------------+--------------+---------------+
| organisasi           | jumlah_lahan | total_luas_ha |
+----------------------+--------------+---------------+
| Gapoktan Tani Makmur |            3 |         2.450 |
| Koperasi Tani Harapan |           2 |         2.600 |
| Gapoktan Mekar Jaya  |            1 |         3.500 |
+----------------------+--------------+---------------+
3 rows in set (0.00 sec)
```

## Q4b — Rekap jumlah catatan per organisasi & jenis (GROUP BY 2 kolom + ORDER BY)

```
SELECT o.name AS organisasi, r.record_type AS jenis, COUNT(*) AS jumlah
FROM organization_records r
JOIN organizations o ON o.id = r.organization_id
GROUP BY o.name, r.record_type
ORDER BY o.name, jumlah DESC;
```

```
+----------------------+----------+--------+
| organisasi           | jenis    | jumlah |
+----------------------+----------+--------+
| Gapoktan Mekar Jaya  | activity |      1 |
| Gapoktan Tani Makmur | activity |      1 |
| Gapoktan Tani Makmur | task     |      1 |
| Gapoktan Tani Makmur | scan     |      1 |
| Gapoktan Tani Makmur | harvest  |      1 |
| Koperasi Tani Harapan | activity |      1 |
| Koperasi Tani Harapan | harvest  |      1 |
+----------------------+----------+--------+
7 rows in set (0.00 sec)
```

## Q4c — Rekap anggota per peran (GROUP BY + COUNT)

```
SELECT m.role AS peran, COUNT(*) AS jumlah_anggota
FROM organization_members m
```

```
GROUP BY m.role
ORDER BY jumlah_anggota DESC;
```

```
+-------------------+----------------+
| peran             | jumlah_anggota |
+-------------------+----------------+
| administrator     |              3 |
| manager           |              2 |
| farmer            |              2 |
| extension_officer |              1 |
+-------------------+----------------+
4 rows in set (0.00 sec)
```

## Q4d — (bonus) rata-rata progres per komoditas (3 tabel + AVG)

```
SELECT c.name AS komoditas, COUNT(r.id) AS jumlah_catatan,
       ROUND(AVG(r.progress_percent), 1) AS rata_progres
FROM organization_records r
LEFT JOIN plots p ON p.id = r.plot_id
JOIN commodities c ON c.id = p.commodity_id
GROUP BY c.name
ORDER BY rata_progres DESC;
```

```
+-----------+----------------+--------------+
| komoditas | jumlah_catatan | rata_progres |
+-----------+----------------+--------------+
| Padi      |              3 |        100.0 |
| Jagung    |              1 |        100.0 |
| Tomat     |              1 |        100.0 |
| Cabai     |              1 |         60.0 |
| Kopi      |              1 |         45.0 |
+-----------+----------------+--------------+
5 rows in set (0.00 sec)
```

## Slot tangkap layar milikmu (opsional — bukti transkrip sudah lengkap)

[Screenshot: konsol mysql saat menjalankan mapan-tugas1.sql]

[Screenshot: hasil Q1–Q4 dari console mysql / MySQL Workbench]

## Kesimpulan

- Sistem informasi MAPAN direpresentasikan sebagai 6 tabel berelasi (≥ syarat 5 tabel) dengan relasi master-detail (organisasi→lahan→catatan) dan N:N (anggota organisasi).
- DDL/DML tereksekusi nyata pada MySQL 8.4.3: **exit 0 tanpa ERROR**; seluruh tabel terisi ≥ 5 baris data simulasi; UPDATE/DELETE terbukti mengubah/menghapus data.
- 8 query pengujian menampilkan, menggabungkan (JOIN 2–3 tabel), memfilter (WHERE), dan merekapitulasi (GROUP BY + COUNT/SUM/AVG) data sesuai kebutuhan sistem.

Disusun dengan bantuan AI coding tool (*vibe coding*) lalu diverifikasi dengan eksekusi nyata. Transkrip penuh: `hasil-eksekusi.txt`; pelan pengerjaan: `docs/plans/planning-tugas-personal-1-database.md`.